Advanced Digital Design Quiz

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QUIZ-1

1. The main advantage of CMOS logic over TTL logic is:

A) Faster switching speed
B) Low power consumption
C) Higher output voltage
D) Requires a higher power supply

2. What does a tri-state buffer do?

A) Adds two signals
B) Amplifies a signal
C) Controls the output between HIGH, LOW, and HIGH IMPEDANCE (Z)
D) Converts analog signals to digital

3. A CMOS XNOR gate requires how many minimum PMOS and NMOS transistors:

a) 8 & 8
b) 6 & 6
c) 9 & 9
d) 5 & 5

4. What is clock skew?

A) Delay in data processing
B) The difference in arrival times of the clock signal at different parts of a circuit
C) The error in a clock pulse
D) Variation in frequency of a clock

QUIZ

5. Industrial automation uses which TTL subfamilies.

a) 74, 74L

b) 74, 74S

c) 74S, 74LS

d) 74AS, 74F

6. The main difference between a PLA (Programmable Logic Array) and PAL (Programmable Array Logic) is:

a) PLA has a programmable AND and OR array, whereas PAL has a fixed OR array
b) PAL has both AND and OR arrays programmable
c) PLA is faster than PAL
d) PAL uses only NOR gates

7. Which of the following is an advantage of using FPGAs?

a) Cannot be reprogrammed
b) Low power consumption but slow speed
c) High flexibility and reprogrammability
d) Only used in military applications

8. In given function, $F = \sum(0, 1, 5, 7, 15, 14, 10)$, find number of EPI, RPI and SPI.

a) 2, 0, 1
b) 3, 1, 4
c) 2, 0, 4
d) 2, 4, 0

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9. The 2's complement form of -68 is represented as:
a) 10111100
b) 10011001
c) 10111101
d) 10101010

10. Which of the following statements is true for an encoder?
a) The number of output lines is greater than the input lines
b) It requires a clock signal to function
c) It reduces multiple inputs into a smaller number of outputs
d) It is the same as a multiplexer

11. Full adder circuit can be designed by using how many minimum nos. of NAND gates.
a) 4
b) 8
c) 9
d) 15

12. The main disadvantage of RCA is:
a) High power consumption
b) Slow carry propagation
c) Large area requirement
d) Complex design

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13. A CMOS XOR gate requires:
a) Only 2 transistors
b) At least 4 transistors
c) More than 6 transistors
d) No transistors

14. The main advantage of CLA over RCA is:
a) Lower power consumption
b) Faster addition due to reduced carry propagation delay
c) Less complexity
d) Requires fewer transistors

15. The Carry Save Adder (CSA) is mainly used in:
a) Multiplication circuits
b) Subtraction circuits
c) Memory circuits
d) ALU of microprocessors

16. Which method is best suited for minimizing Boolean expressions with more than 6 variables?
a) Karnaugh Map
b) Quine-McCluskey Method
c) Truth Table
d) Binary Adder

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17. The Quine-McCluskey method is also known as:
a) K-map method
b) Tabular method
c) Tree method
d) Truth table method

18. In the Quine-McCluskey method, terms differing by only one bit are grouped together to form:
a) Prime implicants
b) Essential prime implicants
c) Minterms
d) Don’t care conditions

19. Essential prime implicants in the Quine-McCluskey method are:
a) Always included in the minimized expression
b) Always excluded from the minimized expression
c) Selected randomly
d) Not important

20. Shannon’s expansion theorem expresses a Boolean function in terms of:
a) Only AND operations
b) Only OR operations
c) A chosen variable and its complement
d) Exclusive-OR (XOR) operations

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21. Shannon's theorem helps in:

a) Expanding Boolean expressions
b) Reducing relay circuits
c) Increasing circuit size
d) Increasing the number of minterms

22. In relay logic, a parallel connection of switches represents which logic operation?

a) AND
b) OR
c) NOT
d) NAND

23. The main disadvantage of the Quine-McCluskey method is:

a) It cannot handle large Boolean functions
b) It does not work with don't-care conditions
c) It becomes computationally expensive for large functions
d) It is only useful for relay circuits

24. What is a hazard in digital circuits?

a) A permanent error in the circuit
b) A short circuit between two logic gates
c) An unwanted fluctuation in the output due to different signal propagation delays
d) A power surge in the circuit

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25. Which of the following best describes a glitch in digital circuits?

a) A permanent logic high output
b) A momentary incorrect output due to hazards
c) A slow response of a logic circuit
d) A malfunction due to incorrect power supply

26. Which type of circuit is more prone to hazards?

a) Combinational circuits
b) Sequential circuits
c) Static RAM
d) ROM

27. In a static-1 hazard, the output:

a) Incorrectly changes from $1 \rightarrow 0 \rightarrow 1$
b) Incorrectly changes from $0 \rightarrow 1 \rightarrow 0$
c) Remains constant
d) Becomes undefined

28. The most common way to eliminate static hazards in combinational circuits is by:

a) Adding redundant logic gates
b) Increasing propagation delay
c) Using a higher voltage power supply
d) Increasing the clock frequency

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29. Which of the following will NOT help in hazard elimination?
a) Adding redundant logic gates
b) Using synchronous clock signals in sequential circuits
c) Increasing the clock speed arbitrarily
d) Using proper logic design techniques

30. Which digital logic family is best suited to avoid hazards?
a) TTL (Transistor-Transistor Logic)
b) CMOS (Complementary Metal-Oxide-Semiconductor)
c) ECL (Emitter-Coupled Logic)
d) DTL (Diode-Transistor Logic)

31. In VLSI design, hazard elimination is important because:
a) It increases propagation delay
b) It improves reliability and performance
c) It increases circuit power consumption
d) It increases the number of transistors

32. Which sub-family of TTL has the lowest power consumption?
a) Standard TTL
b) Low Power TTL (LPTTL)
c) Schottky TTL (S-TTL)
d) Advanced Schottky TTL (AS-TTL)

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33. Why does ECL have high-speed operation?

a) It operates at very high voltage

b) It avoids transistor saturation

c) It uses diodes instead of transistors

d) It consumes low power

34. CMOS logic is widely used because of its:

a) Low power consumption

b) High noise margin

c) High speed

d) Both a & b

35. Which of the following is TRUE about the comparison of TTL, ECL, and CMOS?

a) ECL has the highest speed but consumes more power

b) CMOS consumes more power than TTL

c) TTL has the lowest noise immunity

d) CMOS is the fastest logic family

36. The three states in a tristate buffer are:

a) HIGH, LOW, and DISABLED

b) ON, OFF, and ERROR

c) ACTIVE, INACTIVE, and REVERSE

d) ENABLED, DISABLED, and FAULT

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37. A CMOS NOR gate consists of:
a) Two PMOS in series and Two NMOS in parallel
b) Two PMOS in parallel and Two NMOS in series
c) One NMOS and One PMOS
d) Three PMOS transistors

38. The output of a Muller-C element changes only when:
a) All inputs are HIGH
b) All inputs are LOW
c) All inputs become the same (either HIGH or LOW)
d) One input is HIGH, and the other is LOW

39. What is a key advantage of the Muller-C element?
a) High power consumption
b) Eliminates race conditions in asynchronous circuits
c) Requires a clock signal
d) Works only with synchronous logic

40. Which of the following is true about AOI logic?
a) It cannot be implemented using CMOS
b) It reduces transistor count in circuit design
c) It increases circuit complexity
d) It is slower than basic logic gate design

QUIZ

1. B	21. B
2. C	22. B
3. B	23. C
4. B	24. C
5. C	25. B
6. A	26. A
7. C	27. A
8. C	28. A
9. A	29. C
10. C	30. C
11. C	31. B
12. B	32. B
13. C	33. B
14. B	34. D
15. A	35. A
16. B	36. A
17. B	37. A
18. A	38. C
19. A	39. B
20. C	40 . B

THANK YOU